# 12

## Magnetic Fields and Electromagnetism

# 12.3

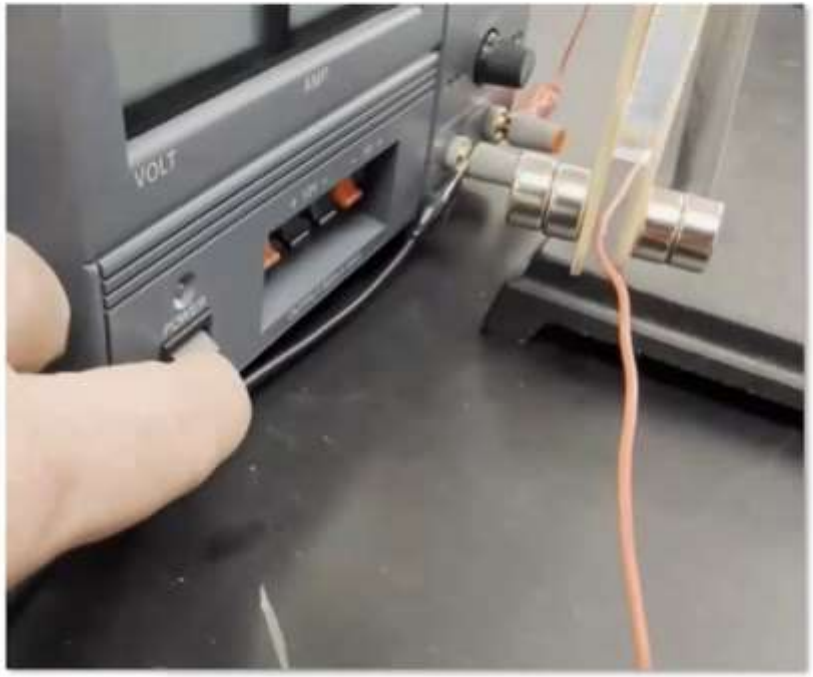
## Magnetic Fields of Current-Carrying Wires and the Biot-Savart Law

# 12.3.1

-Magnetic Forces on Current-carrying wires

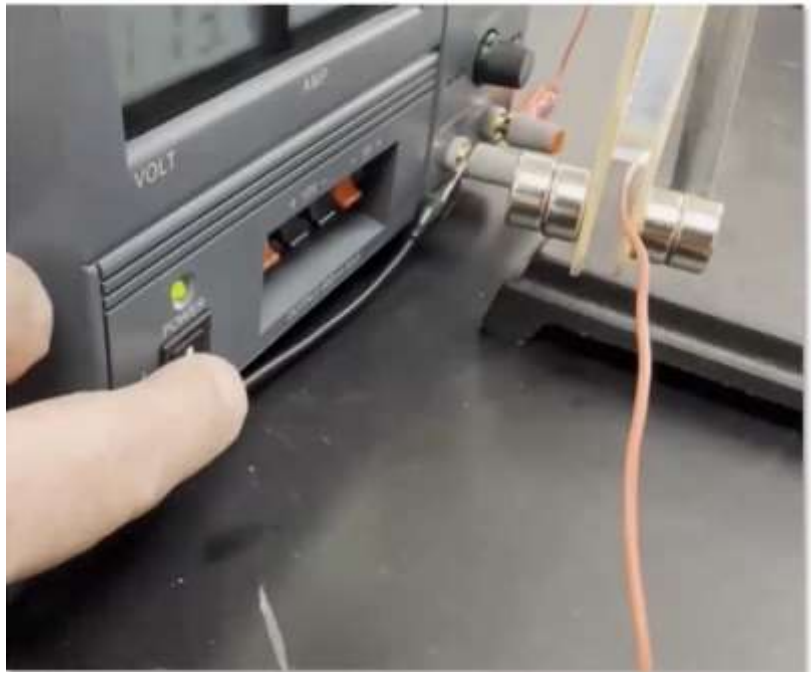
-Changes to the force due to altering current, orientation, or field

## Will a magnetic field exert a force on a current carrying wire?



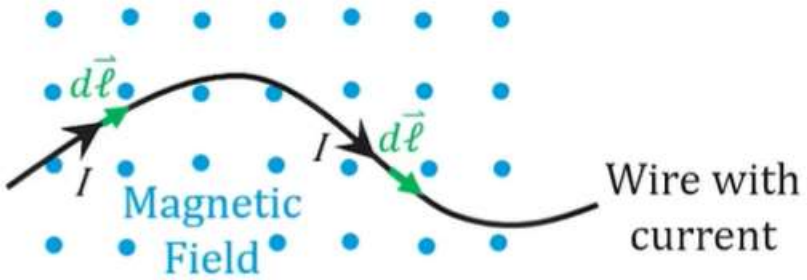

**YES!!!**

You previously learned that moving charges experience a magnetic force, and there are moving charges when current flows.



# Learning Objective: Describe the force exerted on current-carrying wires by a magnetic field.

- $d\overrightarrow{F_B} = I(d\vec{\ell} \times \vec{B})$
- $d\overrightarrow{F_B}$ = Force on small segment of wire $d\vec{\ell}$ due to magnetic field $\vec{B}$
- $I$ = magnitude of current in wire
- $d\vec{\ell}$ = infinitesimally small segment of wire. This should always point the same direction as conventional current flows
- $\vec{B}$ = magnetic field at location of $d\vec{\ell}$

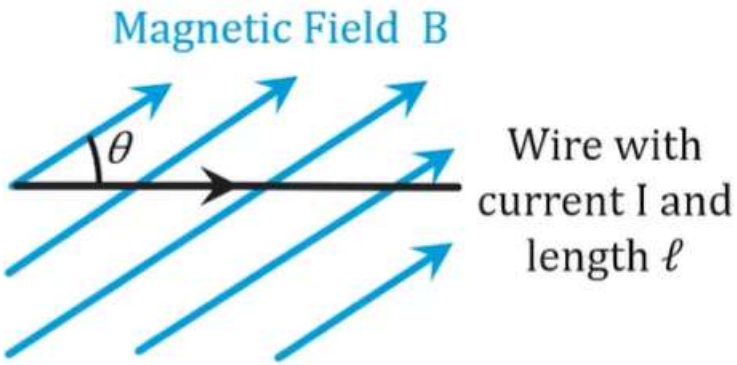



- To find the net force on the wire use calculus to sum up the force on each small segment of wire.

$$\overrightarrow{F_B} = \int I(d\vec{\ell} \times \vec{B})$$

# Let's examine a straight segment of wire with constant current in a uniform magnetic field.

$$\overrightarrow{F_B} = \int I(d\vec{\ell} \times \vec{B})$$




- Simplify cross product $\left|\vec{A} \times \vec{B}\right| = AB \sin\theta$

$$F_B = \int I(d\ell)B \sin\theta$$

- Remove any constants from the integral

$$F_B = IB \sin\theta \int d\ell$$

- Evaluate the remaining integral

$$\boxed{F_B = I\ell B \sin\theta}$$

- Wire must be straight
- Field must be uniform in strength and direction
- Find direction using right hand rule

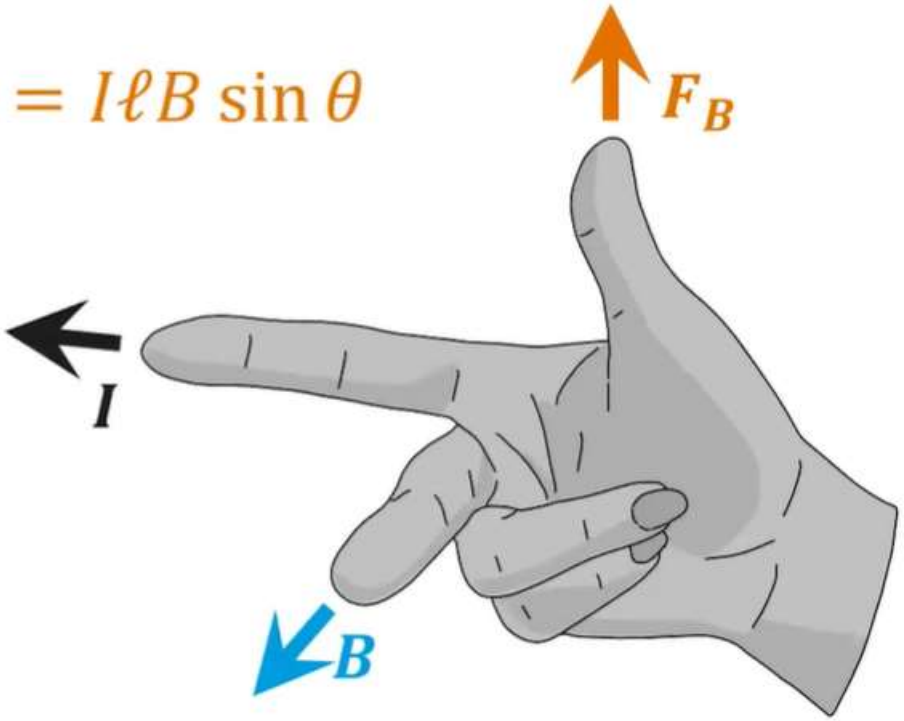
# Review of right-hand rule to find direction of force on straight wire in uniform field.

To find the direction of force…

- Use your right hand!
- Point your index finger in the direction of current ($\boldsymbol{I}$)
- Point your middle finger in the direction of the magnetic field ($\boldsymbol{B}$)
- Your thumb will now point in the direction of force

Note: This will not work if the current and magnetic field are in the same or opposite directions ($\theta$ = 0° or 180°).

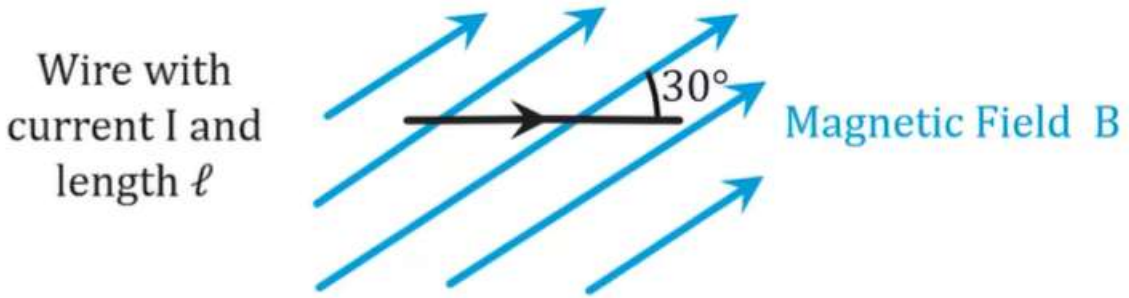
$$F_B = I\ell B \sin\theta$$



# Let's PRACTICE!

# Predict the magnitude and direction of the force in each situation based on the example.

$$F_B = I\ell B \sin\theta$$



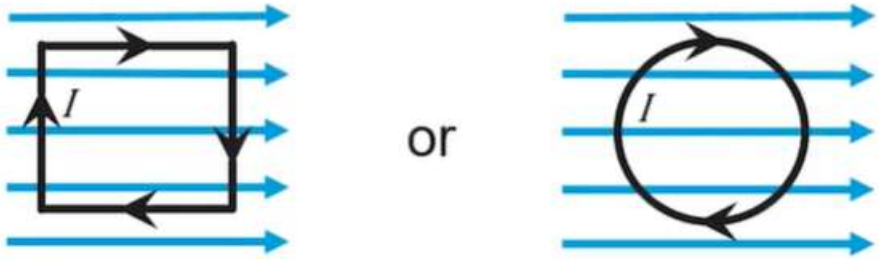

1) If the magnetic field is increased by a factor of 3, the length is cut in half and the angle changed to 60°, by what factor will the force change?

2. If the angle is changed to 90° and the length remains constant, what can be changed to keep the magnitude of force on the wire the same?

3) If the force on the wire increases, and the current decreases, what else might have happened?

# Takeaways

- $\overrightarrow{F_B} = \int I(d\vec{\ell} \times \vec{B})$ is a general equation that can be used for any type of current carrying wire in a magnetic field. This is on the equation sheet.
- There are many types of problems that were not covered in this video, such as finding the torque on a loop of wire in a magnetic field.




- The formula $F_B = I\ell B \sin\theta$ is only for a straight wire in a uniform magnetic field (strength and direction of field must not change). This is *not* on the equation sheet.
- The direction of force is determined using the right-hand rule. There are several versions of the right-hand rule so find one that makes sense to you and know how to use it correctly.

# 12.3.2

Magnetic Field produced by a current-carrying wire, as well as its strength and direction

# Claim: A current carrying wire will produce a magnetic field.

**Question:** Can you design a quick experiment using equipment commonly found in a high school physics laboratory that could verify this claim?

- What equipment would you need?
- What would you do with it?
- How would you know if a magnetic field was present?

**Equipment**

- Voltage Source
- Resistor (to limit current)
- Wire
- Compass

**Procedure**

1. Place compass near the wire with no current flowing.
2. Generate current and see if the compass needle is deflected.
3. Repeat trying different orientations.

# Claim: A current-carrying wire will produce a magnetic field.

**Position 1**

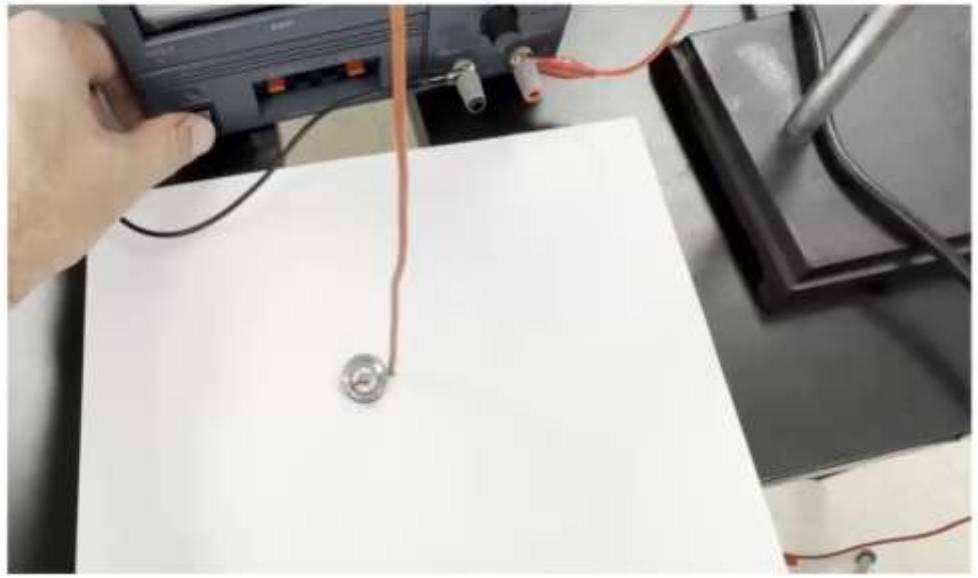
No evidence wire creates magnetic field.

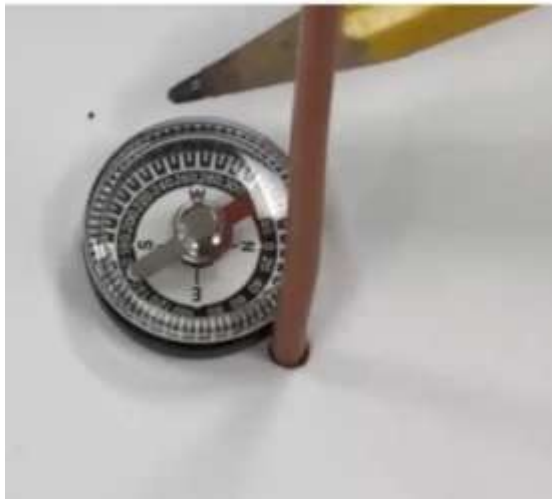
**Position 2**

Compass detects magnetic field of wire.

Note: these claims are tested in the video. Left side shows no deflection in the compass needle, right does.

**Claim: A current-carrying wire will produce a magnetic field forming concentric circles around the wire.**

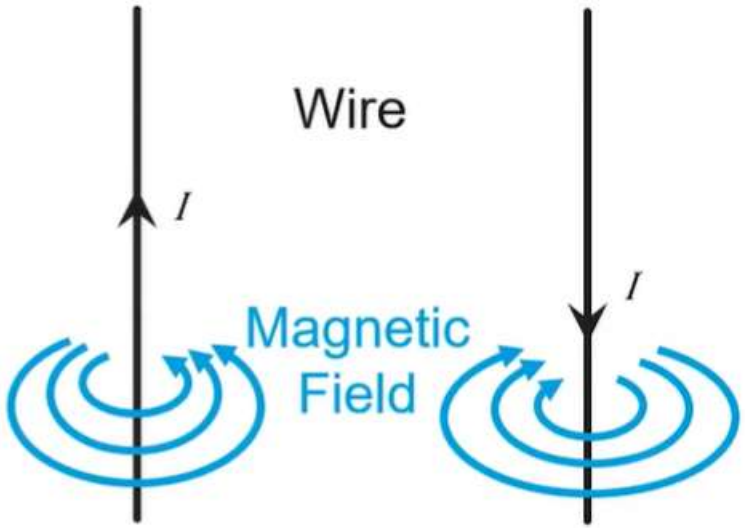


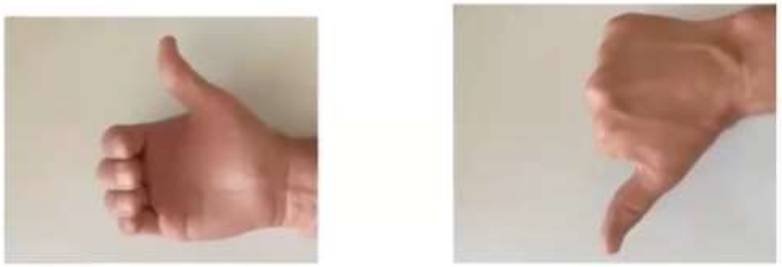


# Direction of the magnetic field around a wire

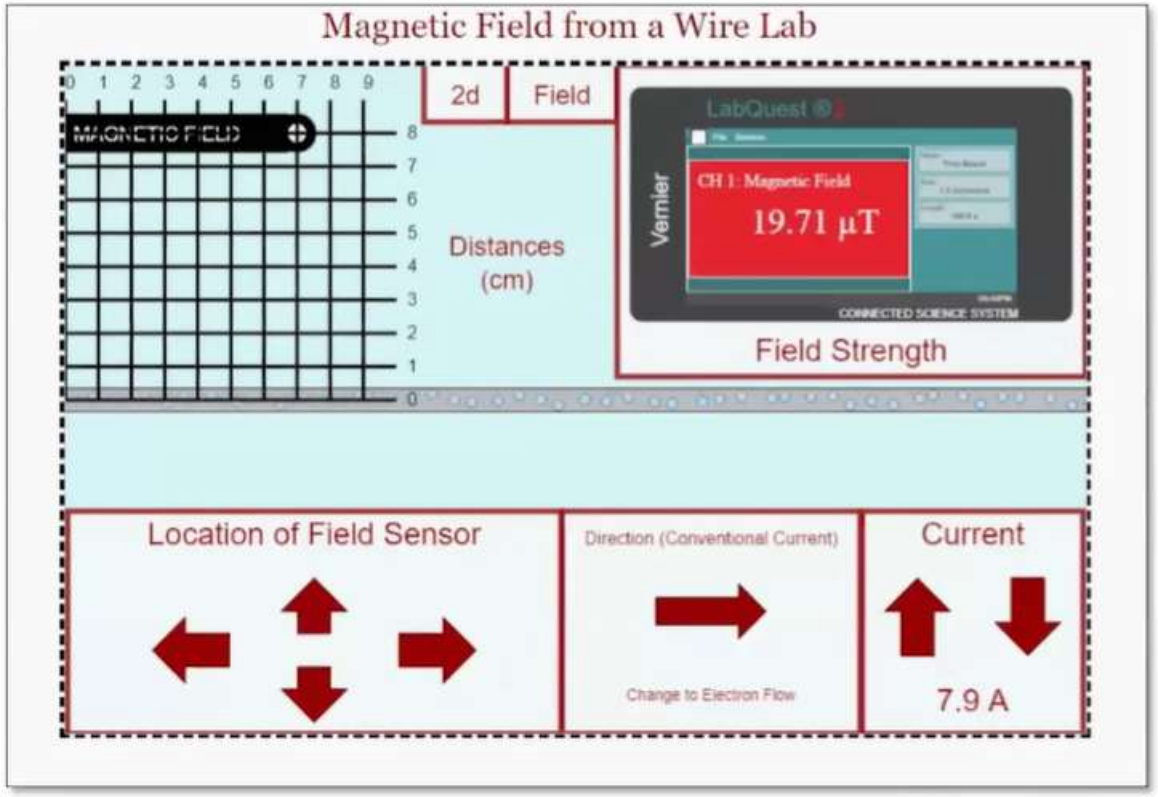
You can use your right hand to predict the direction of the magnetic field created by a current-carrying wire.

- Point your right thumb in the direction of the current.
- The fingers of your right hand will now curl in the direction of the current.



**Claim: The magnetic field created by a long, straight, current-carrying wire will be directly proportional to the current and inversely proportional to the distance from the wire.**

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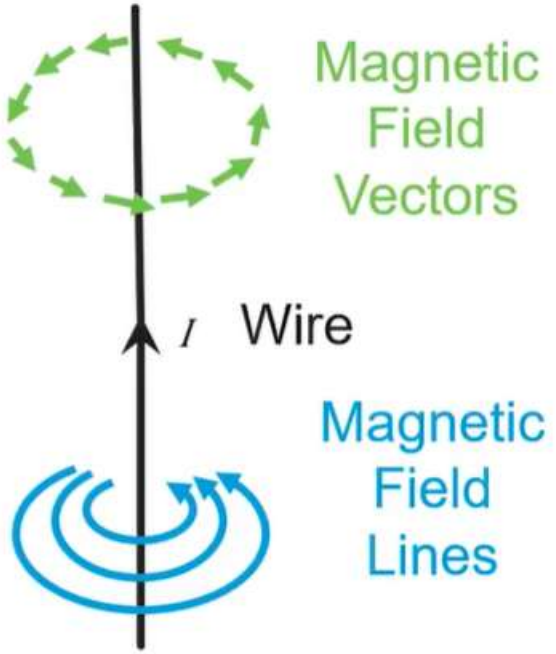
| Current in wire (A) | Magnetic Field Strength ( µT) |
|---|---|
| 0 | 0 |
| ≈ 2 | ≈ 40 |
| ≈ 4 | ≈ 80 |
| ≈ 8 | ≈ 160 |

| Distance from wire (grid lines) | Magnetic Field Strength (µT) |
|---|---|
| 1 | ≈ 160 |
| 2 | ≈ 80 |
| 4 | ≈ 40 |
| 8 | ≈ 20 |

# Takeaways

- Given the idea that moving charges create magnetic fields it follows that current flowing in a wire will produce a magnetic field.
- The magnetic field lines are concentric circles (closed loops) centered on the wire. The magnetic field vectors are tangent to the field lines.
- The field has no component toward, away from or parallel to the segment of wire creating the field.
- The direction of the magnetic field can be found by a variation of the right-hand rule.
- The field grows stronger with more current, and weaker as you move away from the wire.

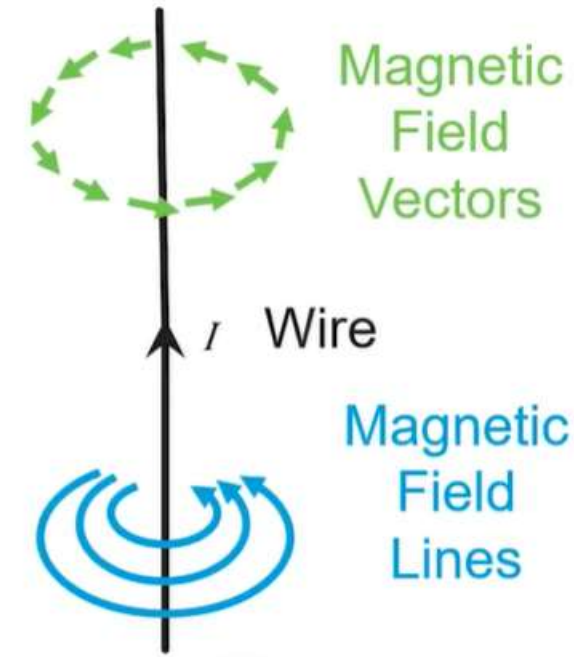


# 12.3.3

Using the Biot-Savart law to derive the magnetic field of a long, straight, current-carrying wire at the center of a current carrying loop

# What did we learn previously?

- Current flowing in a wire will produce a magnetic field.
- The magnetic field lines are concentric circles (closed loops) centered on the wire. The magnetic field vectors are tangent to the field lines.
- The field has no component toward, away from or parallel to the segment of wire creating the field.
- The direction of the magnetic field can be found by a variation of the right-hand rule.
- The field grows stronger with more current, and weaker as you move away from the wire.



# Biot Savart Equation

Used to find the magnetic field of an infinitesimally small segment of a current-carrying wire.

$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(d\vec{\ell} \times \hat{r})}{r^2}}$$

Can be integrated to find…

- the magnetic field of a long straight current-carrying wire.
- the magnetic field at the center of a single loop of current-carrying wire.

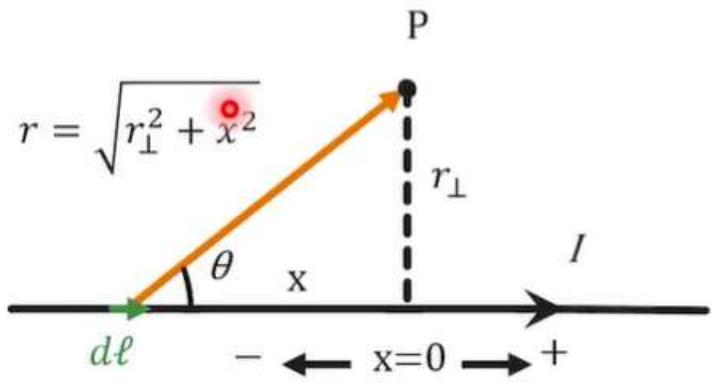
$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(d\vec{\ell} \times \hat{r})}{r^2} \quad \xrightarrow{\text{becomes}} \quad \vec{B} = \frac{\mu_0}{4\pi}\int \frac{I \sin\theta \, d\ell}{r^2}}$$

# Derive the magnetic field of a long straight wire.

Biot Savart Law $\longrightarrow$ $d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(d\vec{\ell}\times\hat{r})}{r^2}$ On equation sheet!

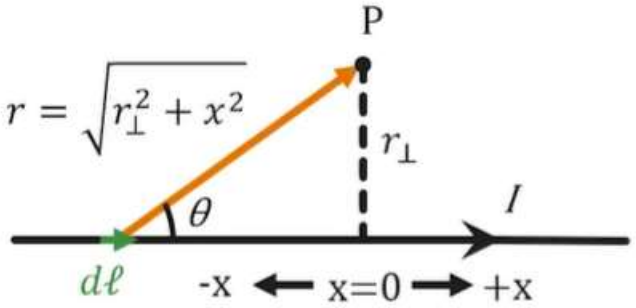
Integrate both sides to get $\left|\vec{B}\right| = \frac{\mu_0}{4\pi}\int\frac{I\sin\theta d\ell}{r^2}$

Now let's apply that to a long straight wire:




- Current ($I$) is constant $\longrightarrow$ remove from the integral.
- $r^2 = r_\perp^2 + x^2$
- $\sin\theta = \frac{opp.}{hyp.} = \frac{r_\perp}{\sqrt{r_\perp^2+x^2}} = \frac{r_\perp}{(r_\perp^2+x^2)^{\frac{1}{2}}}$
- $d\ell = \mathrm{dx}$
- Integrate from $x = -\infty\ to\ x = +\infty$

# Derive the magnetic field of a long straight wire.




$$r^2 = r_\perp^2 + x^2$$

$$\sin\theta = \frac{r_\perp}{(r_\perp^2+x^2)^{\frac{1}{2}}}$$

Don't Panic! This problem is much easier to do using Ampere's Law and will be covered again in a later video.

$$\left|\overrightarrow{B_P}\right| = \frac{\mu_0}{4\pi}\int \frac{I\sin\theta\, d\ell}{r^2}$$

Current ($I$) is constant and removed from integral.

$$= \frac{\mu_0 I}{4\pi}\int \frac{\sin\theta}{1}\frac{d\ell}{r^2}$$

Substitute for $\sin\theta, d\ell$ and $r^2$

$$= \frac{\mu_0 I}{4\pi}\int_{-\infty}^{+\infty} \frac{r_\perp}{(r_\perp^2+x^2)^{\frac{1}{2}}}\frac{dx}{(r_\perp^2+x^2)}$$

Combine the bottoms of the fractions into a single expression

$$= \frac{\mu_0 I}{4\pi}\int_{-\infty}^{+\infty} \frac{r_\perp dx}{(r_\perp^2+x^2)^{\frac{3}{2}}}$$

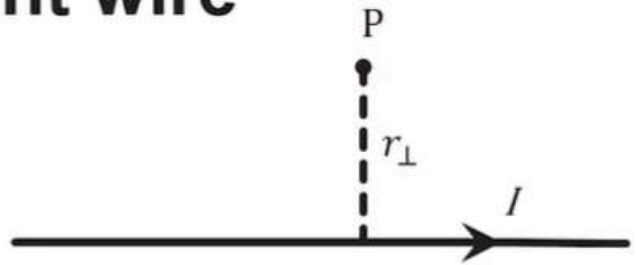
Perform integration and evaluate from $x = -\infty \; to \; x = +\infty$

$$= \frac{\mu_0 I}{4\pi}\left.\frac{r_\perp x}{r_\perp^2(r_\perp^2+x^2)^{\frac{1}{2}}}\right|_{-\infty}^{+\infty}$$

$$= \frac{\mu_0 I}{4\pi}\left(\frac{r_\perp}{r_\perp^2} - \frac{-r_\perp}{r_\perp^2}\right) = \frac{\mu_0 I}{4\pi}\frac{2}{r_\perp} = \frac{\mu_0 I}{2\pi r_\perp}$$

# Magnetic field of a long straight wire

$$\left|\overrightarrow{B_P}\right| = \frac{\mu_0 I}{2\pi r_\perp}$$



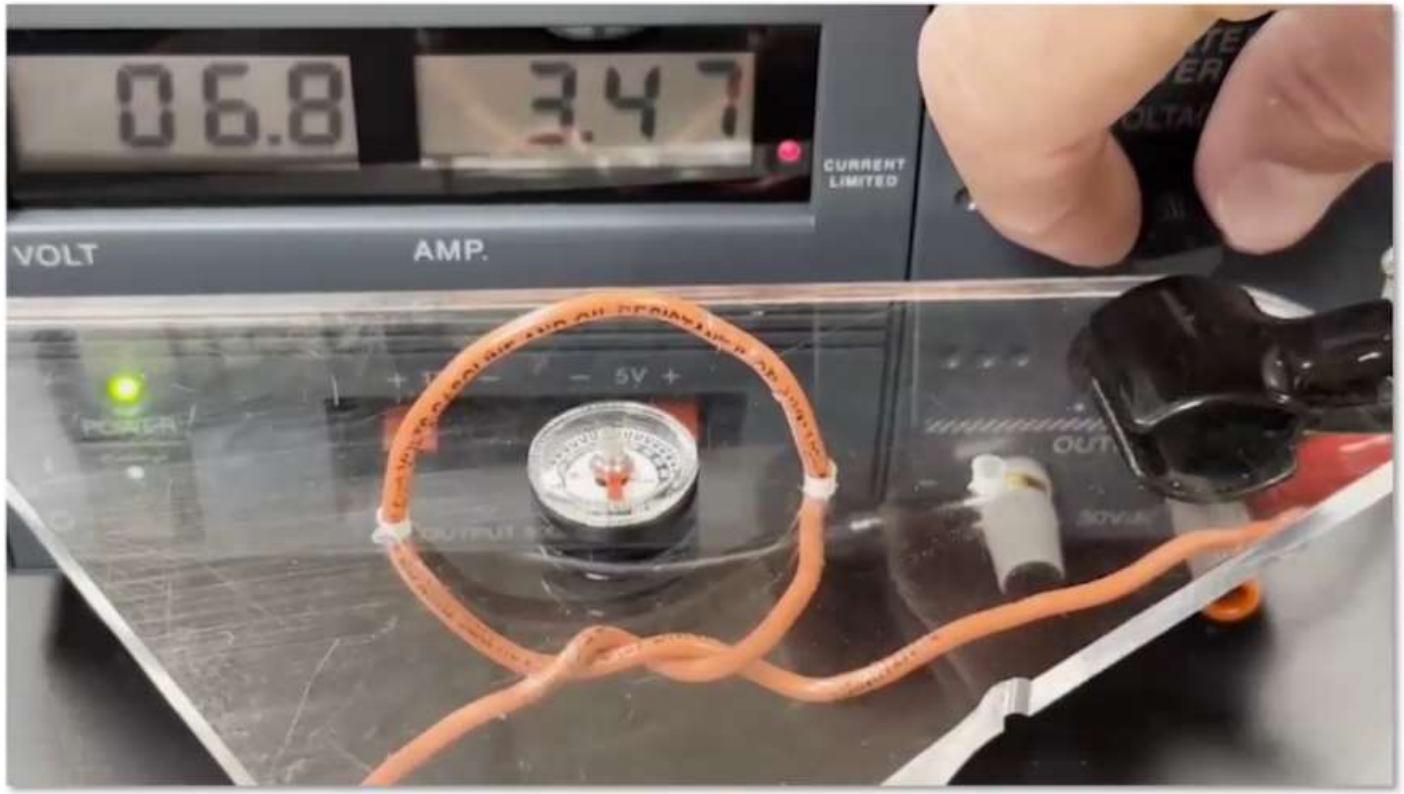

- The magnetic field of a long straight wire is directly proportional to the current in the wire and inversely proportional to the distance from the wire.
- Direction of magnetic field at point P is found by using a variation of the right-hand rule.
- Point thumb in direction of current and fingers curl in direction of circular concentric field lines.
- Magnetic field is out of the page at point P in this example

# Magnetic field at the center of a current-carrying loop

Compass points north when no current flowing.

Compass changes direction in response to the current.

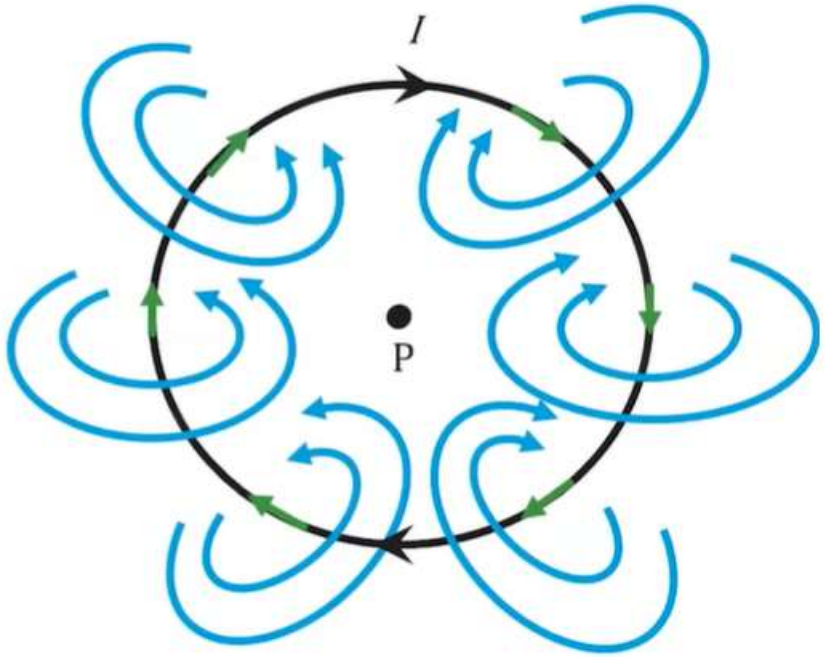
At the center of the loop, the magnetic field created is parallel to the axis of the loop.




Video by Matt Stanich

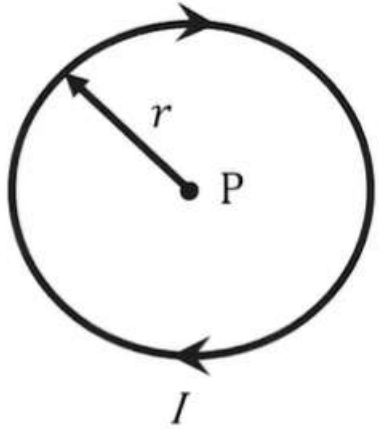
# Magnetic field of a single loop of wire

- Each tiny section of wire creates its own magnetic field.
- These will add together to create the net magnetic field as described by the principle of superposition.
- In the center the magnetic field from each tiny element will be equal due to symmetry
- Use calculus to find the net magnetic field at the center.
- The direction can be found with your right hand. For a loop you can curl your right-hand fingers in the direction of the current, your thumb will point in the direction of the magnetic field.



# Derive the magnetic field of a single loop of a current-carrying wire at the center of the loop.

$$|\vec{B}| = \frac{\mu_0}{4\pi}\int \frac{I \sin\theta \, d\ell}{r^2}$$




<u>Now let's apply that to a single loop of wire:</u>

- Current ($I$) is constant ⟶ remove from the integral.
- Distance from wire to point P ($r$) is constant ⟶ remove from the integral.
- The angle between the radius and the loop is always 90° so $\sin\theta = 1$

$$|\overrightarrow{B_P}| = \frac{\mu_0 I}{4\pi r^2}\int d\ell$$

When you integrate all the $d\ell$ elements you get the circumference ($2\pi r$).

$$|\overrightarrow{B_P}| = \frac{\mu_0 I}{4\pi r^2} 2\pi r = \frac{\mu_0 I}{2r}$$

# Takeaways

- The Biot Savart Law $d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(d\vec{\ell}\times\hat{r})}{r^2}$ is on the equation sheet but must be applied to any specific problem.
- There are problems you may encounter that are not covered in this video.
- Common uses of the Biot Savart Law are…

Magnetic Field of a long straight wire

$$|\vec{B_P}| = \frac{\mu_0 I}{2\pi r_\perp}$$

Magnetic Field of a single circular loop of wire

$$|\vec{B_P}| = \frac{\mu_0 I}{2r}$$

- These equations are **not** on the equation sheet.
- Magnetic field is directly proportional to current and inversely proportional to distance from wire in both cases.
- Ampere's Law (future video) is an easier way to solve the straight wire problem.
- Biot Savart law is the only way to solve for the field at the center of a single loop.